

The US FDA’s Landmark Guidance: Advancing the Use of Artificial Intelligence for Regulatory Decision-Making in Drug and Biological Products

Fahimeh Mirakhori, Independent Consultant (Mirakhori.ff@gmail.com)

Abstract: The technopolitical dynamics of artificial intelligence (AI) are reflected in discourses that underscore its transformative power and disruptive potential. This technology has reached a pivotal stage where regulatory agencies have actively developed guidelines to adopt AI into regulatory decision-making. The United States (US) has consistently led global AI policy and innovation efforts, maintaining dominance in AI investments and regulatory advancements. The US Department of Health and Human Services (HHS) has issued dozens of guidance documents. The US FDA has taken a significant step by releasing its first comprehensive draft guidance on AI-driven regulatory decision-making. In this commentary, we analyze this draft guidance and highlight many areas requiring further clarification, expansion, and alignment to ensure its effectiveness in supporting the evolving needs of the drug industry. We encourage readers to participate in the FDA’s public comment process and actively submit their comments on this guideline to help shape this draft guidance into a globally adaptable model, fostering harmonization across regulatory frameworks. Our analysis highlights the potential of this draft guidance to not only standardize AI adoption in regulatory decision-making but also to set a precedent for international collaboration. By addressing critical gaps and areas for improvement, stakeholders can contribute to creating a robust and forward-thinking framework that advances innovation while ensuring patient safety and regulatory rigor.

Keywords: Artificial Intelligence (AI), Machine Learning (ML), US FDA; Drug Regulatory; Global Harmonization.

1. Introduction

Artificial Intelligence (AI) is revolutionizing drug discovery and development, offering innovative solutions to complex challenges in the pharmaceutical industry [1, 2]. AI technologies, including machine learning (ML) and deep learning, are being applied across various stages of the drug development process, from target identification to clinical trials [3, 4]. The U.S. Food and Drug Administration (FDA) has received numerous product registration submissions relying on AI/ML, prompting the development of stringent guidelines to ensure safe and effective use [5]. AI can potentially expedite drug discovery, enhance drug safety profiles, and improve manufacturing compliance [6]. While AI offers significant opportunities, challenges such as data quality, interpretability, and regulatory concerns persist [7]. Despite these challenges, AI is transforming the pharmaceutical landscape, with implications for personalized medicine and the expansion of therapeutic options for difficult-to-treat diseases [8].

Regulatory agencies worldwide are developing healthcare AI guidelines to ensure safety and effectiveness. The FDA has authorized nearly 1000 AI-enabled medical devices and received numerous submissions for AI-assisted drug development [9]. The agency proposes a total product lifecycle approach for AI regulation, focusing on quality standards and safety [10]. The European Union (EU) also implements AI regulations across sectors, including medical devices [11]. Regulatory initiatives address AI applications in medical devices, post-marketing surveillance, and clinical development [12]. Key considerations include AI systems' transparency, accountability, and interpretability [11]. The FDA emphasizes the need for flexible mechanisms to keep pace with AI advancements and advocates for a lifecycle management approach with post-market monitoring [9]. As

AI transforms healthcare, regulatory frameworks are evolving to balance innovation with patient safety [5, 13-15].

The US FDA has started 2025 with a landmark guidance draft, issued On January 6th, as its first definitive guideline titled “Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products: Guidance for Industry and Other Interested Parties,” [16]. Remarkably, this groundbreaking guidance is sponsored by cross-office collaboration of all major divisions of the FDA, including the Center for Drug Evaluation and Research (CDER), Center for Biologics Evaluation and Research (CBER), Center for Devices and Radiological Health (CDRH), Center for Veterinary Medicine (CVM), Oncology Center of Excellence (OCE) Office of Combination Products (OCP), and Office of Inspections and Investigations (OII), a rare guidance affecting the entire FDA function. It is also directed toward the industry and to all interested parties, creating one of the broadest scopes of any guidance issued by the FDA.

The recommendations in this guidance focus on using AI to produce data or information to support regulatory decision-making for drugs or combination products that include a drug. The recommendations also may be relevant across all medical products, including to support regulatory decision-making for medical devices intended to be used with drugs. It is noteworthy that the European Medicines Agency (EMA) and the Heads of Medicines Agencies (HMAs) published an AI work plan for 2028 in December 2023, setting out a collaborative and coordinated strategy to maximize the benefits of AI to stakeholders while managing the risks [17]. However, the FDA guideline, which is now subject to comments by the stakeholders, is the first comprehensive presentation of the implementation of AI in regulatory decision-making for drug and biological products. We highlight key strengths, including its structured risk-based credibility assessment framework and emphasis on transparency, while identifying areas needing refinement, such as interoperability and the role of real-world evidence (RWE). We offer actionable insights to enhance the utility of this guidance, ensuring that it will fully address the complexities of modern drug development and regulatory sciences. We also recommend that readers consider submitting their views and comments to the FDA to make this model policy more useful for all stakeholders and to adopt it as a model policy for other regulatory agencies.

2. Terminology

This FDA guidance provides recommendations to sponsors and other interested parties on using AI to produce information or data intended to support regulatory decision-making regarding drug safety, effectiveness, or quality. However, to ensure that the terminology used in this guideline is suitably referenced, the guidelines list these definitions for reference purposes that are recommended for consultation by the readers.

As used in this guidance, the term “AI” refers to a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments [18]. AI systems use machine- and human-based inputs to perceive real and virtual environments, abstract such perceptions into models through analysis in an automated manner, and use model inference to formulate options for information or action [18]. A subset of AI that is commonly used in the drug product life cycle, as outlined in this guidance, is machine learning (ML). ML refers to a set of techniques that can be used to train AI algorithms to improve performance at a task based on data [18, 19]. Although ML is currently the most utilized AI modeling technique in the drug product life cycle, this guidance focuses on AI models more broadly.

The FDA guidance provides a risk-based credibility assessment framework that may be used to evaluate and establish the reliability of AI models for specific contexts of use (COU) in regulatory submissions. This framework ensures AI applications align with the severity of potential consequences, prioritizing safety. It focuses on systematic risk management, addressing AI’s impact on patient outcomes while balancing innovation with regulatory rigor. The FDA highlights a high-risk scenario where the AI model stratifies patients based on their likelihood of life-threatening adverse events. This example underscores the critical need for rigorous validation in such applications [16].

Terms like “fit for use” and “context of use” (COU) require more precise definitions to ensure consistent interpretation and application across diverse AI models. “Credibility” refers to a trust established through collecting credibility evidence in performing an AI model for a particular COU. Credibility evidence is any evidence that could support the credibility of an AI model output for a specific COU. The COU defines the distinct role and scope of the AI model used to address a question of interest. This guidance does not endorse using any specific AI approach or technique.

3. Scope of the Framework

The use of AI/ML in drug discovery generally has a lower regulatory oversight if its performance does not affect regulatory review outcomes. However, when AI/ML-driven results contribute to evidence for regulatory decisions, sponsors must ensure that all models and data sets are thoroughly assessed for any potential risks. This assessment is crucial for risk mitigation plans, including biases and ethical concerns focusing on data quality and quantity during data collection and model training [1-11, 64].

While this guidance covers many broad areas related to regulatory decision-making, it does not address the use of AI models in drug discovery or when used for operational efficiencies (e.g., internal workflows, resource allocation, drafting/writing a regulatory submission) that do not impact patient safety, drug quality, or the reliability of results from a nonclinical or clinical study [16]. To clarify, the FDA recommends sponsors engage with the FDA early if they are uncertain whether their use of AI is within the scope of this guidance. By excluding drug discovery and operational efficiencies in the guideline, its utility is significantly compromised since AI is the most likely to impact these two areas. This can introduce risks, including compromised data quality and integrity, leading to unreliable predictions and unsafe biologics productions. Ethical concerns, such as bias in AI models, may result in products that are less effective or safe for diverse populations. Moreover, non-compliance with regulatory standards can delay market approval and hinder product development. To mitigate these risks, it is essential that AI models and datasets used in these domains be reviewed for ethical issues and data integrity. Therefore, special attention should be paid to the processes of data acquisition and augmentation to ensure integrity and minimize bias.

The “risk-based credibility assessment framework” is one of the main foci to help sponsors and other interested parties plan, gather, organize, and document information to establish the credibility of AI model outputs when the model is used to produce information or data intended to support regulatory decision-making. FDA emphasizes that the activities such as the level of oversight by the FDA, the sponsor, or other parties responsible for the relevant information or data, the stringency of the credibility assessments and the performance acceptance criteria, the risk mitigation strategy, and the type and extent of documentation and detail associated with AI use, used to establish the credibility of AI model outputs should be commensurate with the AI model risk and tailored to the specific COU.

FDA offers several options by which sponsors and other interested parties may engage with the Agency on issues related to AI model use, depending on the COU and the specific development program.

While the guidance emphasizes data reliability and mentions limitations and risks, it offers limited actionable strategies for addressing biases in AI models. While biases are acknowledged, actionable steps to identify, quantify, and mitigate these biases—especially those affecting underrepresented groups—remain insufficient.

The FDA promotes early engagement to integrate innovative technologies while upholding high regulatory standards. Sponsors must provide detailed documentation, including the AI model's development, rationale for use, limitations, biases, and maintenance processes, ensuring transparency and trust in regulatory decisions. This early communication helps set clear expectations, address potential challenges, and allows sponsors to refine their models based on feedback, reducing the risk of delays or rejections. These

proactive measures foster regulatory clarity and build confidence in the decision-making process.

The proposed extensive lifecycle maintenance and validation requirements may disproportionately impact more minor sponsors or those with limited resources. Provisions to support such entities would improve accessibility. Smaller companies may face challenges in meeting the extensive documentation and validation requirements. Offering tailored guidance or support for more minor sponsors could enhance accessibility. Tailored guidance or fee adjustments could improve accessibility.

The scope of the FDA guidance primarily focuses on risk-based credibility assessments, yet it does not fully address the broader implications of quality assurance (QA) in AI/ML applications. The QA roles evolve from traditional, retrospective data checks to real-time, AI-driven insights. This shift could significantly enhance the scope of the guidance by embedding QA considerations at every phase of the drug development lifecycle, strengthening the guidance's relevance and utility.

One key gap is the lack of emphasis on QA practices for ensuring data accuracy and consistency through AI model development and deployment. Effective QA is essential to maintaining the reliability and validity of AI-driven processes, especially as these systems are applied to high-stakes decision-making in drug development.

Additionally, the guidance does not provide a clear framework for addressing the evolving roles of QA in AI/ML applications, particularly for multinational sponsors who must navigate diverse regulatory landscapes. This omission limits the guidance's applicability globally, where alignment with international standards is crucial.

To address these gaps, we recommend broadening the scope of the guidance to explicitly include QA as a core component of AI/ML systems. This expansion should emphasize QA from data collection and preprocessing to model deployment and real-time monitoring. Incorporating AI-driven QA processes would not only ensure data reliability but also streamline regulatory documentation and compliance tasks. Furthermore, aligning the guidance with global standards would enable multinational sponsors to adopt consistent QA practices, reducing risks and fostering greater harmonization. By highlighting the role of AI in enhancing QA roles, the FDA can encourage more precise and efficient regulatory processes, ensuring that AI/ML are leveraged effectively while safeguarding patient outcomes. The FDA guidance should incorporate a modular approach to AI regulation to address these gaps, emphasizing QA across the entire product lifecycle. Introduce dynamic QA mechanisms that adapt to evolving AI systems, such as real-time monitoring tools, self-healing algorithms, and autonomous test orchestration models. These tools would enable continuous assurance and minimize risks associated with self-learning algorithms [20].

Traditional QA struggles with incomplete, inaccurate, or biased datasets, impacting model reliability and performance. Manual testing cannot scale effectively to handle large datasets and complex system requirements in modern AI applications. Cognitive bias, fatigue, and inconsistent processes can compromise manual QA efforts. Rapid development cycles and iterative updates introduce difficulties in maintaining consistent QA processes and validation benchmarks. Conversely, AI-driven QA tools create test cases autonomously based on natural language inputs, reducing the time and effort required for manual test creation. Machine learning models analyze datasets to identify and mitigate bias or inconsistencies, ensuring a higher level of accuracy in testing. AI continuously monitors system performance and offers actionable insights, enabling QA teams to address issues during development proactively. AI eliminates human error by standardizing QA processes and adhering to predefined guidelines, enhancing reliability across all testing phases. AI integrates QA into the entire product lifecycle, from development and testing to deployment and post-market monitoring, ensuring consistent performance [21].

Incorporating robust QA strategies is vital for enhancing ML applications' trustworthiness, accuracy, and scalability in big data analytics. It advocates for a shift from traditional QA methods to dynamic, automated, and adaptive approaches that address the unique challenges posed by ML systems. We recommend that the FDA enhance the draft

guidance to provide clear guidance on the specific QA strategies sponsors should adopt when developing such AI/ML applications.

While the guidance targets AI's role in regulatory decision-making, it excludes applications related to drug discovery or operational efficiencies not directly tied to safety, effectiveness, or quality evaluations. Expanding the scope to include AI's role in the pre-clinical and discovery phases could enhance its utility. Expanding its scope could better integrate AI throughout the drug development lifecycle. In addition to AI guidance, the FDA also issued recommendations on accelerated approvals [22, 23] and using tissue biopsies [24] in clinical trials, reflecting its multifaceted approach to modernizing drug development processes.

While the guideline indirectly acknowledges AI's role in improving patient care through optimized clinical trials and accelerated approvals, it fails to address patient equity in AI models. Expanding the framework to incorporate Patient Preference Information (PPI) and address accessibility issues and user fees could significantly enhance patient-centric outcomes. PPI submissions, which highlight trade-offs patients are willing to make, could help AI models prioritize risks and benefits better. Transparent AI systems considering patient preferences may also build trust and engagement, while streamlined AI-driven processes could reduce costs and improve timely access to innovations. Additionally, clarity on user fees would aid more minor sponsors in navigating regulatory complexities, ultimately benefiting patients. We recommend fostering early and frequent collaboration with regulators, payers, and health technology assessments (HTAs) to align methodologies and expectations. In rare disease contexts, prioritizing unmet patient needs through AI could ensure that real-world preferences are effectively integrated into decision-making.

3.1. Real World Evidence

The guidelines need to emphasize the necessity of using real-world data (RWD) and evidence (RWE) with modern analytical techniques to better regulatory decision-making. The 21st Century Cures Act required the FDA to explore RWE to speed medical product development. RWE uses real-world data from Electronic Health Records (EHRs), claims, registries, and surveys. Rare diseases, vaccines, and oncology are increasingly regulated using RWE [19, 25-27].

RWE has excellent potential, but many obstacles prevent its full integration into regulatory decision-making. Interoperability difficulties, irregular EHR coverage, and data quality inconsistencies might limit reliable analysis. Governance and privacy are significant difficulties because they limit data reuse and regulatory approval. RWE must be translated into credible evidence using approved statistical methodologies and solid analytical frameworks to ensure dependability and regulatory applicability. Addressing these issues is necessary to maximize RWE's medication development and regulatory benefits.

The FDA guidance emphasizes RWD in regulatory decision-making and clinical trials but does not reference external control arms (ECAs). ECA refers to the use of data from tools outside the randomized clinical trial, such as from previous trials, observational studies, or RWE sources, to serve as a comparison group for the experimental treatment. The recommendation stresses using AI to analyze large datasets and improve trial efficiency while ECA integration is still under consideration.

Per the FDA's proposed guidance, AI models can be assessed throughout the drug development process, including nonclinical, clinical, and manufacturing phases. ECAs in clinical trials demonstrate RWD's capacity to improve trial data and regulatory submissions. International initiatives emphasize the need for well-defined criteria to help regulatory bodies embrace ECAs, but the FDA's guidance doesn't cover the complexities of ECA implementation. The guiding and RWD/RWE techniques share information, showing how AI and ECAs can transform drug development. Both stress the need for (1) comprehensive regulatory frameworks for innovative techniques and (2) harmonization of global RWD, RWE, and AI decision-making standards.

The FDA's proposed guidance focuses on AI model validation and risk-based frameworks, not statistical methods or interoperability issues. The shortcomings highlight the opportunity to integrate pragmatic trials, Bayesian modeling, and EHR interoperability into RWE frameworks and AI-informed regulatory decision-making.

The FDA guidance addresses AI-specific issues, and integrating advanced analytical methods into RWE frameworks may help close gaps and unify decision-making. AI can effectively process and analyze large datasets from pragmatic trials and interoperable electronic health record systems to generate valuable insights, improving real-world evidence. RWE ensures that AI models are trained on real-world data, improving generalizability and reliability. AI-RWE integration in both directions could improve regulatory decision-making and medication development.

3.2. Real World Evidence

AI and RWE have advanced regulatory decision-making with the FDA's draft guidance. However, adjustments can improve its application and efficacy. The guidance can help drug development's changing landscape by addressing advanced methodology integration, interoperability, and RWE framework alignment gaps.

The FDA's draft guidance is a significant step in integrating AI and RWE into regulatory decision-making. However, several enhancements can strengthen its applicability and effectiveness. The guidance can better serve the evolving landscape of drug development by addressing gaps in integrating advanced methodologies, interoperability, and alignment with RWE frameworks.

First, the guidance could explicitly incorporate pragmatic trial methodologies, such as adaptive designs and patient-centered approaches, and evaluate their contribution to drug development into its framework [19, 28]. Pragmatic trials are designed to evaluate interventions in real-world settings and can bridge the gap between clinical trial data and real-world outcomes. Their incorporation into the guidance would provide a pathway for using RWE generated from diverse populations to validate AI models and assess their performance in practical healthcare environments. This alignment would enhance the credibility of AI applications while ensuring that the data used reflects real-world conditions, especially in rare diseases and complex treatments.

A second critical area of improvement is the integration of Bayesian modeling into the FDA's credibility assessment framework [19], (see section 6; steps 3-6). Bayesian methods provide a dynamic approach to data analysis, allowing for integrating prior knowledge and new data to refine decision-making [29]. This methodology is particularly valuable in scenarios with limited data, such as small patient populations or rare disease studies for genomics discovery and clinical predictions. By explicitly recommending Bayesian approaches, the FDA could offer sponsors a robust statistical method to address uncertainty, improve generalizability, and strengthen the evidence base for regulatory decisions.

In drug development and clinical trials, ensuring data quality and model generalizability is critical for making informed regulatory decisions. Data quality metrics focus on evaluating the completeness, accuracy, consistency, and representativeness of the dataset. Completeness ensures all necessary data points are present, while accuracy focuses on the precision of data labels, measurements, and annotations. Consistency is essential for uniform data collection methods, and representativeness ensures the dataset mirrors the target population, avoiding bias in model predictions. Model generalizability metrics such as cross-validation, external validation, sensitivity analysis, and out-of-sample prediction help assess whether the model can perform effectively on unseen data, ensuring the results can be trusted in diverse clinical or real-world settings (Table 1).

Bayesian models effectively address data quality and generalizability by integrating prior knowledge and estimating uncertainty, which is crucial for decision-making with incomplete or noisy data. These models are widely used in regulatory contexts, such as the FDA's integration of Bayesian methods in clinical trials and data from diverse sources

like EHRs. Bayesian techniques refine models iteratively in adaptive trials, improving decision-making as new data becomes available [29-30].

By combining Bayesian models with RWE, regulators ensure more reliable, data-driven decisions, especially when traditional trials are not feasible. Bayesian methods also enhance model generalizability, adapting to diverse populations and limited datasets, ensuring more robust regulatory outcomes.

Incorporating data quality metrics like completeness, accuracy, and consistency, Bayesian models adjust for low-quality data, improving predictions and decision-making over time. This iterative process refines models, supporting better evaluations of new drugs and medical devices, particularly in AI-driven approaches.

Table 1. Data quality metrics and model generalizability [16, 19, 64].

Area	Data Quality Metrics	Model Generalizability	How the New FDA Guidance Draft Can Improve	Relevant Section in FDA Guidance Draft
Data Integrity & Accuracy	Accuracy of data labels (e.g., diagnosis, treatment outcome)	Use of diverse datasets that represent different populations (e.g., age, sex, ethnicity)	The FDA can emphasize the importance of data accuracy by including mandatory validation checks in the model development phase.	Data Quality and Verification (Step 2)
Bias Detection & Mitigation	Evaluation for bias in the dataset (e.g., sampling bias, measurement bias)	Ensure the AI model performs well across a wide range of subgroups and diverse populations (demographics)	Expand guidance to include strategies for identifying and mitigating bias in datasets to ensure the model is generalizable to all patient groups.	Data Quality and Verification (Step 2)
Data Sources & Provenance	Source verification of data (e.g., electronic health records, clinical trials, insurance claims)	Model should be trained on data from various real-world sources (e.g., registries, EHR, RWD)	Guidance could specify the importance of incorporating RWE RWD from multiple sources to ensure broad applicability.	Real-World Evidence and Data Use (Step 3)
Data Completeness	Completeness of datasets (e.g., missing values, completeness of patient records, trial data)	Ensuring model's performance is evaluated on datasets from multiple clinical settings, regions, and time periods	FDA can recommend methods for data imputation or exclusion criteria, and emphasize the importance of ensuring dataset completeness before model development begins.	Data Collection, Monitoring, and Auditing (Step 4)
Data Preprocessing & Cleaning	Standardization of data formats, normalization, and cleaning to remove outliers and noise	Testing model performance on independent datasets (external validation)	Expand the draft to provide clear guidelines on data preprocessing methods, particularly for data sourced from real-world scenarios (e.g., EHRs, RWE, RWD).	Data Collection and Quality Control (Step 4)

External Validation	Use of external validation datasets to test model performance in non-training environments	Evaluate model performance across multiple datasets to test its adaptability to new, unseen data	FDA guidance could stress the necessity of using external validation datasets as a critical step to assess the model's generalizability across different datasets.	Model Evaluation and Validation (Step 5)
Cross-validation	Cross-validation techniques to estimate model performance on various subsets of the data	Adapting models to new environments or scenarios (e.g., different hospitals, regions, patient groups)	Strengthen the guidance to encourage cross-validation as a robust method to assess model reliability and generalizability across multiple patient populations.	Model Evaluation and Validation (Step 5)

Bayesian statistics offer a flexible and adaptive framework for decision-making in pharmacological studies, early drug development, clinical trials, and post-approval by integrating prior knowledge with current data. The FDA’s CDER and CBER have recognized the value of Bayesian methods, as evidenced by their guidance on medical device trials. These methods enhance trial design, improve efficiency, and facilitate informed decision-making, leading to shorter development timelines and reduced costs. By integrating Bayesian methods with RWE, regulators gain a more complete understanding of how medical products perform across diverse patient populations, leading to more efficient regulatory processes and faster approvals. For example, the FDA applied Bayesian methods to approve devices like the Multi-frequency Impedance Breast Scanner, which integrated multiple clinical studies to support regulatory decisions [19, 29-30].

In drug development, Bayesian techniques have proven especially useful for evaluating treatments for rare diseases and oncology, where small patient populations complicate the use of traditional trial designs. The integration of Bayesian models with RWE enables more data-driven and informed decisions, enhancing the robustness and reliability of regulatory outcomes. Additionally, Bayesian Neural Networks (BNNs) and EHR interoperability are vital for developing trustworthy AI models in healthcare. These models predict uncertainties in data, improving the accuracy of predictions and enabling timely patient interventions [29, 30].

In the early phases of drug development, Bayesian models can significantly enhance decision-making by allowing researchers to update their understanding of a drug's behavior based on accumulating data, including data from small patient populations. This is particularly valuable when addressing complex dose-response relationships, small sample sizes, or exploring rare diseases where traditional clinical trial methods may not be feasible or efficient. For example, in pharmacokinetic (PK) and pharmacodynamic (PD) modeling, Bayesian techniques enable the integration of prior information from preclinical studies, early clinical trials, and the literature to inform the design of Phase I and II clinical trials. This iterative learning process aids in optimizing dosing schedules, identifying biomarkers, and assessing drug safety and efficacy more efficiently. In pediatric pharmacokinetics, Bayesian modeling has been used to predict drug dosing in children based on data from adults, adjusting for age-related differences in drug metabolism. This has been particularly useful in drugs like antiretrovirals for HIV treatment, where pediatric clinical trials are difficult to conduct due to the limited patient population [30-35].

One notable example is the use of Bayesian adaptive designs in early-stage clinical trials, where the model is continually updated as data are collected, allowing modifications to the trial design during the study. This approach can reduce development time, trial burden, and sample size, lower costs, and enhance decision-making, particularly in complex therapeutic areas like oncology or rare diseases. The use of Bayesian models

permits sponsors to explore a wider range of hypotheses, diminish uncertainty, and make more informed decisions about which compounds or doses to advance, thereby accelerating drug development. This combination is particularly valuable in scenarios where traditional randomized controlled trials (RCTs) may be impractical. Therefore, we recommend that the scope of the guidance expands to include the integration of Bayesian models and RWE to enhance AI-driven decision-making processes, particularly in rare diseases, urgent clinical contexts, and when traditional clinical trials are difficult to implement.

The guidance must also address the challenges of EHR interoperability, which is critical for ensuring the quality and accessibility of real-world data [36]. Interoperability enables seamless data exchange between healthcare systems, ensuring that AI models are trained on diverse and representative datasets. Without clear interoperability standards, data quality and coverage inconsistencies may compromise the credibility of AI outputs. The FDA should outline requirements for interoperability and collaborate with stakeholders to develop frameworks that support cross-platform data sharing. This would enhance the reliability of AI applications and enable more efficient use of RWE in regulatory contexts.

Finally, the guidance should work toward bridging the gap between AI and RWE frameworks. While the current draft focuses on AI-specific challenges, a more integrated approach is needed to maximize AI's and RWE's utility. For example, AI can analyze large, complex datasets generated by pragmatic trials or interoperable EHR systems, providing actionable insights that enrich RWE. Conversely, RWE can inform the development and validation of AI models, ensuring they are trained on data reflective of real-world scenarios. By explicitly addressing how these frameworks intersect and complement each other, the FDA can create a cohesive regulatory strategy that leverages the strengths of both approaches. Incorporating pragmatic trials, Bayesian modeling, and EHR interoperability into the guidance will ensure that AI models are credible, reliable, and reflective of real-world applications. Furthermore, aligning AI and RWE frameworks will create a unified approach to regulatory decision-making, enabling the FDA to adapt to the complexities of modern drug development. By addressing these areas, the FDA can set a global standard for the ethical, effective, and innovative use of AI and RWE in advancing patient care.

4. Global Trials

The growing global acceptance of innovative tools, including AI-driven methods, underscores their importance in enhancing clinical trials and regulatory submissions. While the 21st Century Cure Act guides these advancements in the U.S. through the FDA, agencies like the EMA and China's CDE pursue similar modernization initiatives under their respective regional frameworks and policies. The FDA's guidance does not explicitly address region-specific challenges or provide a detailed comparison of RWE adoption across regions such as the U.S., Europe, and China. However, the draft guidance aligns with global trends, including the EMA's Big Data Task Force and HMA/EMA initiatives [19, 37], which focus on harmonizing international standards for leveraging RWD, RWE, and AI. Emphasizing collaboration with global regulatory agencies to create consistent frameworks, reduce duplication, and ensure equitable adoption across markets is essential for harmonizing with international regulatory frameworks to facilitate broader adoption for multinational sponsors. These three global regions rank among the most active in advancing AI within the life sciences industry, with the United States maintaining its leadership in global AI investments and driving innovation [38]. Despite its protectionist stance and claims of technological superiority, the EU struggles with a smaller digital industry and lower investments than global powers like the United States and China.

The FDA's framework must align with international efforts (e.g., EMA's initiatives, Health Canada's policies, China's CDE) to facilitate global AI adoption and reduce redundancy for multinational companies—the G7 working group's recommendations on adaptive AI oversight and thresholds for regulatory review. The CMDE's Guiding Principles

for AI-Enabled Medical Devices emphasize lifecycle quality control and preemptive risk identification—harmonizing with the EMA’s efforts to standardize European AI regulation [19, 25].

The FDA guidance could provide more specifics on how sponsors should ensure model transparency [19, 39]. Cross-border analyses emphasize adopting ethical AI standards to address biases and data quality and ensure equitable outcomes for diverse populations. Globally, regulators prioritize explainability in AI systems to build trust among users, regulators, and patients. By aligning with global frameworks, the FDA can facilitate smoother regulatory pathways for multinational sponsors and promote consistent safety standards worldwide. Furthermore, these encouraging regulatory and policy frameworks emphasize the need for rigorous QA practices in critical sectors such as clinical data management. They should aim to establish standardized adaptive QA frameworks that evolve alongside data and models. Tailored for AI/ML-driven real-time monitoring and controls, such frameworks can optimize testing processes, minimize human intervention, and enhance trust and accountability through explainability mechanisms [21]. Additionally, they must ensure consistency across the lifecycle to maintain reliability and transparency.

5. Drug Product Lifecycle Advancements

In recent years, the use of AI in the drug product life cycle has increased continuous advancements in AI potential to accelerate the development of safe and effective drugs and enhance patient care. Concurrent with these technological advancements, the use of AI in regulatory submissions to the FDA has also increased for some uses [40].

Some examples [41] of AI uses for producing information or data intended to support regulatory decision-making regarding the safety, effectiveness, or quality of drugs include:

- Reducing the number of animal-based pharmacokinetic, pharmacodynamic, and toxicologic studies, using predictive modeling for clinical pharmacokinetics and/or exposure-response analyses,
- integrating data from various sources (e.g., natural history, clinical studies, genetic databases, clinical trials, social media, registries) to improve understanding of disease presentations, heterogeneity, predictors of progression, recognition of disease subtypes;
- processing and analyzing large sets of data (e.g., data from real-world data sources or data from digital health technologies) for the development of clinical trial endpoints or assessment of outcomes;
- identifying, evaluating, and processing for reporting post-marketing adverse drug experience information; and
- facilitating the selection of manufacturing conditions.

However, AI use presents some unique challenges. First, the variability in the quality, size, and representativeness of datasets for training AI models may introduce biases and raise questions about the reliability of AI-driven results. As such, data used to develop AI models should be fit for use, which means the data should be both relevant (e.g., includes key data elements and enough representative participants or sufficient data that is representative of the manufacturing process or operation) and reliable (i.e., accurate, complete, and traceable) [42].

Second, because of the complex computational and statistical methodology underpinning these models, understanding how AI models are developed and how they arrive at their conclusions may be difficult and necessitate methodological transparency (e.g., detailing in the regulatory submission the methods and processes used to develop a particular AI model).

Third, uncertainty of the accuracy in the deployed models’ output may be difficult to interpret, explain, or quantify. Finally, another challenge with some AI models is the potential for the model’s performance to change over time or across deployment environments when new data inputs are introduced, and these inputs differ from the data on

which the model was trained (i.e., data drift) requiring life cycle maintenance of these models.

Human subjects' protections are out of the scope of this guidance but should be considered when developing or deploying AI modeling in the drug product life cycle, as applicable. For additional information [43].

The FDA's guidance emphasizes model development and maintenance but lacks detailed provisions for real-time monitoring of AI outputs during clinical trials and post-market phases. For high-risk applications, such as patient stratification in life-threatening conditions, additional guidance on real-time monitoring and fallback mechanisms is necessary to ensure safety and prevent unintended consequences. The guidance does not explicitly address continuous assurance protocols, leaving unclear when significant changes from self-learning mechanisms should trigger a new regulatory review. This gap increases the risk of performance degradation going unnoticed.

Traditional regulatory models evaluate safety and functionality at predefined points in a product's lifecycle, requiring incremental updates supported by data. While effective for static systems, these models are less suited to managing the adaptive and iterative nature of AI/ML-enabled technologies, which can evolve autonomously through self-learning mechanisms. While such systems improve real-time optimization, they also raise concerns about diminished human oversight, unforeseen risks from data adaptation without reevaluation, and ethical issues arising from biased or poorly labeled data.

In the same recommendation as given for other elements, the guidance encourages early and continuous engagement with the FDA, setting expectations for ongoing evaluation and transparency. Sponsors are urged to document and validate AI updates, highlighting the importance of lifecycle maintenance. These recommendations partially align with a Total Product Lifecycle Approach (TPLA) advocated for adaptive AI technologies [19, 44].

The draft guidance provides an important foundation for using AI/ML in drug development but does not adequately explore its transformative role in QA processes. AI is revolutionizing QA roles by automating error detection, enhancing process efficiency, and improving data accuracy in real time. These capabilities are crucial in clinical trials and manufacturing, where ensuring data quality directly impacts regulatory decision-making. These include insufficient exploration of AI's role in automating QA processes, such as identifying data inconsistencies, flagging anomalies, and ensuring traceability. Limited focus on the proactive capabilities of AI to enhance QA workflows, particularly in detecting errors earlier in the trial process or during manufacturing, can be very troublesome. Discussing AI-driven QA tools and best practices can dynamically monitor and validate clinical trial data, reducing manual errors and inefficiencies. AI-enable QA can integrate seamlessly into existing workflows to reduce costs, improve timelines, and ensure high-quality data reliability for regulatory submissions.

To address these challenges comprehensively, the FDA should incorporate specific continuous assurance mechanisms into its framework [19, 44, 45]. This would include requirements for real-time monitoring of adaptive AI systems post-deployment, criteria for identifying significant changes that necessitate additional regulatory review, and clear thresholds for determining when self-learning updates pose safety or efficacy risks. Such measures would ensure robust oversight while supporting adaptive AI/ML technology innovation.

Moreover, ethical concerns in AI/ML systems often arise from biases in data labeling, poor-quality training datasets, and lack of transparency in decision-making [19]. The ethical concerns extend beyond data quality and labeling issues. These include biases in model training that could lead to skewed predictions, the risk of model hallucination where AI generates misleading or incorrect conclusions, and critical data privacy challenges. Furthermore, the FDA emphasizes the importance of ensuring diversity and equity in clinical datasets, as AI-driven decisions must reflect the broad spectrum of patient populations to prevent perpetuating healthcare disparities and to maintain the credibility of risk assessments in regulatory decision-making.

Traditional oversight models primarily focus on initial evaluations and are inadequate for managing dynamic and evolving algorithms in autonomous AI systems. While the FDA guidance emphasizes model transparency by requiring detailed descriptions of AI models, their rationale, and limitations, it lacks actionable steps for identifying and mitigating algorithmic biases. Furthermore, the ethical implications of using adaptive AI in patient care—where errors can directly impact safety—are not comprehensively addressed.

The guidance's focus on early engagement and ensuring data reliability and quality is foundational to ethical AI development. However, explicit requirements for bias assessments and mitigation plans should be included to strengthen ethical oversight. High-stakes AI applications should mandate the use of human-in-the-loop systems to ensure human oversight remains integral, even in adaptive deployments. Additionally, aligning ethical standards with international frameworks, such as the G7 working group's principles, could foster global consistency and strengthen the ethical foundation of AI systems [19].

Another important consideration is that the quality and representativeness of data used during initial development significantly influence the generalizability of AI models [19]. Poor data quality often results in models that perform well in controlled environments but fail in diverse real-world settings. While the FDA guidance highlights the need for reliable fit-for-purpose data representative of the intended application, it does not provide specific metrics or methods for assessing generalizability during product development. Additionally, there is insufficient emphasis on ensuring data diversity to prevent biases disproportionately affecting underrepresented populations.

To address these gaps, the guidance should include specific recommendations for assessing data diversity and generalizability [19]. This could involve defining minimum requirements for data representativeness across demographics, geographies, and disease states and validation protocols to test AI performance in real-world scenarios. Encouraging external datasets for independent validation would further enhance the robustness of AI models and help identify potential generalizability issues before deployment. These steps ensure that AI models deliver consistent and equitable performance across real-world conditions.

6. Step-by-Step Guide for the Risk-Based Credibility Framework

Among various computational models used in the drug product life cycle, this guidance focuses on using AI models to produce information or data intended to support regulatory decision-making regarding drug safety, effectiveness, or quality.

The risk-based credibility assessment framework described here consists of the following 7-step process to establish and assess the credibility of an AI model output for a specific COU based on model risk. This structured approach provides actionable steps for ensuring AI models are robust, reliable, and fit for their intended regulatory contexts (Figure 1).

Moreover, the seven steps outlined in the FDA guidance draft for the risk-based credibility framework can be aligned closely with the Quality Management System (QMS) principles and Lean Six Sigma's methodology. The FDA's framework mirrors DMAIC's phases (Define, Measure, Analyze, Improve, Control) by defining the AI model's purpose, measuring and analyzing associated risks, developing improvement plans, and establishing controls to ensure the model's reliability and compliance throughout its lifecycle. This alignment underscores a shared focus on quality, safety, and efficiency.

In today's rapidly evolving digital era, AI's ability to adapt and self-learn introduces non-deterministic testing capabilities, enabling QA processes to grow dynamically as new data emerges. This capability aligns with the FDA's emphasis on risk-based assessments and model credibility, as outlined in its seven-step framework for AI model validation and context of use (COU) assessments. However, while the guidance emphasizes data quality, transparency, and lifecycle monitoring, it lacks detailed protocols for fully

leveraging AI in QA, particularly in data annotation, test environment optimization, and autonomous error correction (i.g., steps 3 and 4).

The current FDA guidance draft does not adequately integrate QA considerations and actionable steps into its risk assessment framework. To address these gaps, we recommend mandating the inclusion of comprehensive QA protocols in the credibility assessment plans. Rather than treating QA as isolated checkpoints, the focus should shift to continuous oversight, ensuring model validation, error mitigation, and lifecycle management remain robust. In the FDA draft guidance, limited attention is given to QA protocols that ensure the integrity of AI models during updates, adaptations, or real-world applications.

Furthermore, the guidance does not offer clear recommendations for documenting QA activities as part of the credibility assessment plan, leaving a critical oversight in ensuring transparency and accountability. AI/ML systems generate outputs that are often opaque, making it challenging for stakeholders to interpret their underlying logic. This lack of interpretability could hinder the credibility of QA processes and erode trust among regulators and sponsors. The FDA guidance acknowledges the importance of transparency but does not provide clear requirements for labeling or describing AI systems' inputs, logic, or outputs. Notably, AI/ML-enabled devices that self-learn and evolve autonomously present novel challenges for product liability and accountability. Traditional QA frameworks focus on static, human-designed algorithms, which are inadequate for managing AI systems that revise their algorithms based on RWE. Therefore, the regulatory framework should aim to establish standardized adaptive QA plans that evolve alongside data and models while ensuring consistency across the lifecycle. However, the FDA's guidance does not explicitly address how to assess and regulate these dynamic changes over a product's lifecycle.

Acceptance testing is a crucial phase within the QA process, serving as the final validation to ensure that a system or product meets predefined requirements and is ready for deployment [21]. In the context of AI/ML applications, during acceptance testing, the QA process ensures that the datasets used for training and testing the AI system are complete, accurate, and free of bias. This is critical for AI systems that automate decision-making processes, as errors could compromise product quality. This testing benchmarks the AI system's predictions against ground truth data to validate its performance under various conditions. Precision, recall, and false favorable/negative rates are assessed to ensure reliability. For instance, scenarios where deviations in manufacturing parameters fall outside the model's training data are tested to ensure fallback mechanisms are in place. The acceptance testing results are documented and submitted as part of the regulatory dossier to demonstrate the AI system's robustness and compliance with FDA guidelines. The process includes transparent reporting of the model's limitations and risk management strategies. For regulatory decision-making, such as the FDA's risk-based credibility framework, acceptance testing aligns with Step 4—developing a plan to establish credibility—by confirming that AI models meet performance benchmarks and provide trustworthy outcomes under regulatory conditions.

AI-driven QA accelerates the testing process by automating repetitive tasks, allowing QA teams to focus on strategic decisions and complex problem-solving. This enables broader and deeper testing coverage, identifying edge cases and hidden anomalies that might go unnoticed in traditional testing. AI supports continuous and real-time QA processes, ensuring ongoing validation of software systems and reducing the risks associated with updates or new releases. AI-driven QA evolves with the system, adapting to dynamic software requirements, data models, and new functionalities. Leveraging AI/ML-driven QA systems can proactively identify potential risks by continuously monitoring data integrity and ensuring the robustness of underlying datasets. This approach is particularly critical for high-risk AI applications, such as patient stratification in clinical settings or automation processes in pharmaceutical manufacturing, where inaccuracies can lead to severe consequences for patient outcomes or product quality.

Moreover, Bayesian models are increasingly being used in post-market surveillance to enhance regulatory decision-making by incorporating RWD to monitor drug safety and efficacy over time. One notable example is the use of Bayesian methods in the FDA's REMS (Risk Evaluation and Mitigation Strategies) programs, where Bayesian approaches help analyze post-market safety data to identify potential risks or side effects that might not have been detected in pre-market trials. For example, the FDA employed Bayesian methods to assess the safety of certain drugs in post-market surveillance, adjusting the probability of adverse events as new data from EHRs and other RW sources became available. Additionally, the FDA's use of Bayesian models in pharmacovigilance has facilitated more effective signal detection for adverse drug reactions (ADRs), improving the ability to react to emerging safety concerns in real-time [30-35]. These Bayesian techniques allow for the continuous integration of new data, making the safety monitoring process more dynamic and responsive, ultimately enhancing the FDA's ability to make informed regulatory decisions based on RWE. By adopting Bayesian methods, the FDA can refine post-market risk assessments, ensuring ongoing patient safety while minimizing unnecessary restrictions on drug availability. For example, in pediatric rare diseases such as Duchenne Muscular Dystrophy (DMD), Bayesian modeling and simulations have been used to combine data from various studies and existing literature to estimate treatment effects. The model helps to predict the outcomes of smaller trials, optimizing trial designs and making the most of limited data [34].

Bayesian methods were used in the development of a pediatric version of the rotavirus vaccine. The integration of prior adult data, RWE, and early pediatric data helped in estimating the optimal vaccine schedule for infants, leading to faster approval of the vaccine for use in young children [35].

These examples demonstrate how Bayesian models, through their ability to integrate prior knowledge and adapt to new data, provide a flexible and efficient framework for addressing the unique challenges of pediatric drug development. This approach not only enhances the design of clinical trials but also supports faster and more informed decision-making in the regulatory process. While this approach offers potential for improving the feasibility of pediatric clinical trials, it demands careful consideration of inter-study variability and expert guidance to ensure its effective implementation. We propose that the draft guidance incorporate the use of Bayesian models and RWE into the product life cycle risk assessment framework. This should include post-market surveillance and ongoing monitoring to reflect risk mitigation, adverse effects, and overall product performance, particularly in assessing safety and efficacy across diverse patient populations, for pediatric and rare diseases.

Risk-Based Credibility Framework (FDA Guidance draft)

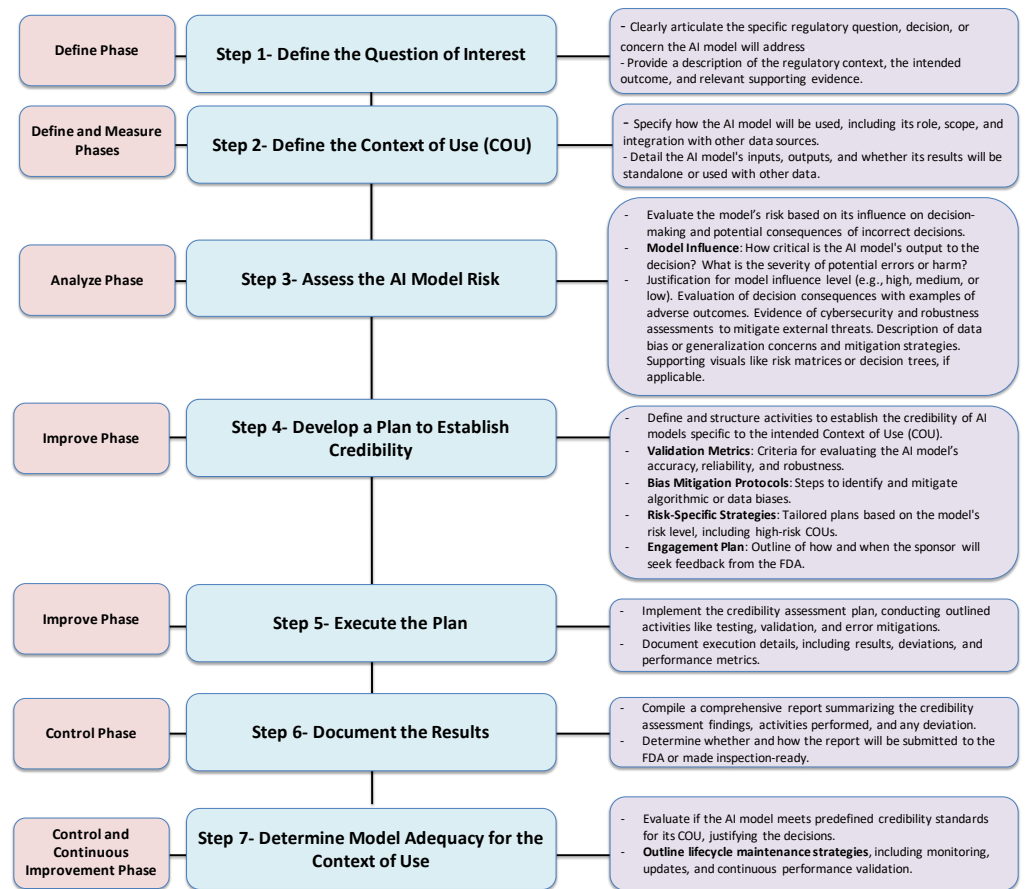


Figure 1. US FDA's Seven-Step Risk-Based Credibility Assessment Framework for AI Models in Regulatory Decision-Making for Drug and Biological Product Development [16].

6.1. Step 1- Define the Question of Interest (Define Phase)

The first step in the framework is to clearly define the question of interest that the AI model will address. This question should describe the specific decision, concern, or objective the model supports. For instance, in the context of clinical development, Drug A is under development and is associated with a life-threatening adverse reaction. Data from previous trials indicate that some participants are at low risk for this reaction, and an AI model is proposed to stratify patients based on their risk levels. The question of interest here is: "Which participants can be considered low-risk and do not need inpatient monitoring after dosing?" Similarly, the fill volume is a critical quality attribute in a manufacturing scenario for Drug B, a parenteral injectable in multidose vials. An AI-based visual analysis system is proposed to assess vial fill levels automatically. The question of interest in this case would be: "Do vials of Drug B meet established fill volume specifications?"

6.2. Step 2- Define the Context of Use (COU) (Define and Measure Phases)

The next step involves defining the AI model's context of use (COU). This includes detailing the specific role and scope of the AI model and how its outputs will be used to address the question of interest. In the clinical development example, the AI model is intended to predict a participant's risk for a life-threatening adverse reaction to Drug A based on baseline characteristics and lab values. This prediction will stratify participants into low- and high-risk groups, determining whether inpatient or outpatient monitoring is required. In the manufacturing example, the AI model analyzes vial images to identify deviations in fill volume. However, the model's outputs will complement existing quality

control measures, such as batch-level volume verification, rather than acting as the sole determinant for product release.

6.3. Step 3- Assess the AI Model Risk (Analyze Phase)

Assessing the risk associated with the AI model is a critical step in the framework. Model risk is determined by two factors: the influence of the AI model in the decision-making process and the potential consequences of incorrect decisions. In the clinical development example, the model influence is high because the AI model is the sole determinant of the monitoring strategy, and the decision consequence is also high due to the potential for life-threatening outcomes if errors occur. Therefore, this model would be classified as high-risk. In contrast, the decision consequence remains high for the manufacturing example since vial fill volume is a critical quality attribute. Still, the influence of the model is reduced by using complementary quality control processes. This results in a medium-risk classification for the manufacturing AI model.

Model risk is the possibility that the AI model output may lead to an incorrect decision that could result in an adverse outcome and not risk intrinsic to the model (The decision consequence is the significance of an adverse outcome resulting from an incorrect decision concerning the question of interest. Decision consequence is the potential outcome of the overall decision that is made by answering the question of interest, outside of the scope of the AI model and irrespective of how modeling is used. That is, decision consequence should consider the question of interest but should not consider the COU of the model. Additionally, when assessing the decision consequence, the FDA recommends that sponsors consider both the potential severity of adverse outcomes and the probability that the adverse outcome would occur. In some risk management tools, the ability to detect harm (detectability) is also a factor in risk estimation. For more information, see the guidance for industry and FDA staff Assessing the Credibility of Computational Modeling and Simulation in Medical Device Submissions (November 2023). Assessing model risk involves subject matter expertise and judgment among sponsors, interested parties, and the FDA [9-11, 30].

This model risk matrix can be applied to the clinical development examples to address the question of interest, such as “Which participants can be considered low risk and do not need inpatient monitoring after dosing?” In this example, model influence would likely be estimated to be high because the AI model will be the sole determinant of which type of patient monitoring a participant undergoes. The decision consequence is also high because if a participant who requires inpatient monitoring is placed into the outpatient monitoring category, that participant could have a potentially life-threatening adverse reaction in a setting where the participant may not receive proper treatment. Given that model influence is deemed high for this question of interest and decision consequence is also considered high, the model risk for this COU is high. Other types of risk, such as cybersecurity risk, are out of the scope of this guidance but should be considered when deploying AI modeling in the drug product life cycle.

For the commercial manufacturing example, deviations in the volume of vials containing the drug could result in several issues. For example, releasing units that do not meet quality standards could lead to medication errors due to either an inability to withdraw labeled content or the pooling of vials to obtain a single dose (if not identified in the labeling) [46]. Since volume is a critical quality attribute and incorrect volume measurements would greatly impact product quality, the decision consequence would be high. However, for this example, a manufacturer would measure fill volume on a representative sample for each batch as a part of release testing. Measuring fill volume through release testing would reduce the influence of the AI model, and therefore, the influence of the model would be determined to be low. Given that the decision consequence is deemed high and the model influence is deemed low with the stated mitigations, the model risk for this COU is medium.

Insufficient incorporation of QA metrics within the risk assessment process undermines key QA dimensions, such as data completeness, accuracy, and annotation consistency, leaving room for variability and potential errors in AI/ML model outputs. Additionally, the guidance lacks robust tools and methodologies for evaluating the quality of training and test datasets, which are essential components in determining model risk and reliability. We recommend introducing QA-specific criteria within the risk assessment framework to address these gaps. These criteria should include data accuracy measures, bias detection mechanisms, and error tolerance thresholds to ensure that AI/ML models operate within acceptable performance parameters. This approach will not only improve patient safety but also streamline regulatory processes, providing higher confidence in the adoption of AI-driven innovations.

Assessing model risk is important because the credibility assessment activities used to establish the credibility of AI model outputs, described in step 4, should be commensurate with the AI model risk and tailored to the specific COU.

6.4. Step 4- Develop a Plan to Establish Credibility (Improve Phase)

Once the model risk and COU are defined, developing a credibility assessment plan is next. This plan outlines the activities needed to establish the reliability of the AI model's outputs. It should include considerations for data quality, model validation, performance metrics, and methods for identifying and mitigating potential biases. The activities in the credibility assessment plan should be commensurate with the model risk. High-risk models require more stringent validation and detailed documentation, while lower-risk models may only require minimal assessments. Early engagement with the FDA during this stage can help refine the plan and set expectations for credibility activities.

The guideline details the general considerations and assessment activities related to establishing and evaluating the credibility of AI model outputs that can be included in such plans. These general considerations and assessment activities are not meant to be exhaustive, and some may not apply to all AI models and contexts of use.

Whether, when, and where the plan will be submitted to the FDA depends on how the sponsor engages with the Agency and on the AI model and COU (The Agency recognizes that specific uses of AI occur outside of contexts with established meeting options. Specifically, in post-marketing pharmacovigilance, specific documentation (e.g., processes and procedures) is not generally submitted to the Agency. Still, it is maintained according to the sponsor's standard operating procedures and made available to the Agency upon request (e.g., during an inspection). In such cases, sponsors may complete all the steps outlined in the guidance without seeking early engagement with the Agency.

For example, the plan could be described in a formal meeting package as detailed by the FDA in several guidelines. [47] [48]. However, sponsors remain responsible for compliance with statutory and regulatory requirements, including post-marketing safety surveillance and reporting requirements, regardless of the technology utilized.

The risk-based credibility assessment framework envisions interactive feedback from the FDA concerning assessing the AI model risk (step 3) and the adequacy of the credibility assessment plan (step 4) based on the model risk and the COU. Accordingly, the FDA strongly encourages sponsors and other interested parties to engage early with the FDA to discuss the AI model risk and the appropriate credibility assessment activities for the proposed model based on model risk and the COU. Although detailed information on all the credibility assessment activities described in the guidance may not be available or necessary to include at the time of early engagement with FDA, the proposed credibility assessment plan about which the sponsor engages with the Agency should, at a minimum, include the information described in steps 1, 2, and 3 (i.e., question of interest, COU, and model risk) and the proposed credibility assessment activities the sponsor plans to undertake based on the results of those steps. In early discussions with the Agency, the proposed credibility assessment activities in the credibility assessment plan might be more high-level, with a more detailed credibility assessment plan drafted after the iterative process.

As noted previously, the potential use of AI in the drug product life cycle is broad and rapidly evolving. Therefore, the activities that may establish the credibility of AI model outputs should generally be tailored to the specific COU and commensurate with model risk. For example, the performance acceptance criteria should be more stringent and described to the FDA in more detail for high-risk models than low-risk models.

6.4.1. Model Description

The sponsor's credibility assessment plan submitted to the FDA for early consultation should include the sponsor's proposed credibility assessment activities based on the question of interest, COU, and model risk. As noted, early descriptions of those activities may be high-level, with further details provided after the agency's feedback. In addition, for specific low-risk models, FDA may request minimal information in the categories described below. For high-risk models, FDA may request all the information in the categories described below and additional information, as applicable, depending on the COU.

Sponsors and other interested parties should include the following information in the credibility assessment plan, as applicable, for each AI model used:

An explanation of each model used to include, but not limited to, descriptions of model inputs and outputs and model architecture (e.g., convolutional neural network) [49, 50]

Features play a role in training and prediction. In the clinical development example, model features include baseline demographic characteristics and lab values for trial participants (adapted from ISO/IEC 23053:2022) [51]. A model feature is a measurable property of an object or event concerning a set of characteristics. Features include:

- Clinical measurements, demographics, and clinical imaging data.
- Feature selection process and any loss function(s) used for model design and optimization, as appropriate
- Model parameters (For this guidance, a model parameter is an internal variable of a model that affects how it computes its outputs. Examples of parameters include the weights in a neural network and the transition probabilities in a Markov model (adapted from ISO/IEC 22989:2022 [52]).

For this guidance, the data used to develop the model are generally composed of training and tuning data (Although some in the AI and ML communities sometimes use the term validation to refer to the tuning data and the tuning process, FDA does not use the word validation in this context.) (collectively, development data) as part of the development stage.

Training data are those used in procedures and training algorithms to build an AI model, including to define model weights, connections, and components. Tuning data are typically used to evaluate a few trained AI models. More than one tuning dataset may be used in the tuning process. The tuning process involves exploring various aspects of model development, including different architectures or hyperparameters. The tuning phase happens before the testing phase of the AI model and is part of the development phase.

The performance of an AI model relies heavily on the datasets used to train and tune the model. Therefore, the data used to develop the AI model should be fit for use, which means the data should be both relevant (e.g., includes key data elements and enough representative participants or sufficient data that is representative of the manufacturing process or operation) and reliable (i.e., accurate, complete, and traceable). The definitions of training and tuning data for this guidance are consistent with those terms discussed in other FDA guidance [53].

Commensurate with model risk, sponsors and other interested parties should describe the data management practices for the development datasets (i.e., training and tuning datasets) and characterize the development datasets. These descriptions may help identify potential data limitations, including potential sources of algorithmic bias. Data management is also important for identifying and mitigating bias and promoting health equity. Algorithmic bias is a possible tendency to produce incorrect results in a systematic,

but sometimes unforeseeable, way due to limitations in the training data or erroneous assumptions in the machine learning process. For example, during training, models can be over-trained to recognize features unique to specific patient subpopulations that have little to do with generalizable patient anatomy, physiology, or condition, which can result in AI bias in the resulting model. Additionally, for example, the underrepresentation of specific populations in datasets could lead to overfitting (i.e., data fitting too closely to the potential biases of the training data) based on demographic characteristics, which can impact the AI model performance in the underrepresented population), and the appropriate credibility assessment activities to support the use of the AI model for a particular COU. Sponsors and other interested parties should include the following information in the credibility assessment plan, as applicable.

Developing and evaluating AI models in drug product life cycles requires meticulous attention to data management and model performance. First, the development datasets must be clearly described, including details on splitting them into training, tuning, and other subsets and specifying which activities are performed using each dataset. Data collection, processing, annotation, storage, and control methods should be thoroughly documented, with a rationale for selecting the datasets. It is essential to explain how labels or annotations are established and how the data are fit for the intended Context of Use (COU), ensuring they are relevant and reliable. Additional considerations include whether the development data are centralized, such as through federated learning, and a description of the specific activities performed using each dataset.

Model training details should address the learning methodology employed (e.g., supervised or unsupervised learning) and the metrics used to evaluate model performance, such as sensitivity, specificity, and F1 scores, all accompanied by confidence intervals. Techniques to prevent overfitting or underfitting, such as regularization, should be discussed, along with training hyperparameters like loss function and learning rate. If a pre-trained model is used, its dataset, development process, and rationale should be specified. The description should include ensemble methods, model calibration for accuracy and repeatability, and quality assurance processes to track software version changes.

The model evaluation process, which assesses performance on independent test data, must ensure data independence from the training and tuning datasets. This requires specifying how independence was achieved and justifying any overlapping data use between the development and testing phases. The evaluation should include a description of the test data's relevance to the COU and methods to address data drift, which could impair the model's real-world applicability. Metrics such as the receiver operating characteristic (ROC) curve, sensitivity, specificity, and other relevant indicators must be documented, and methods to estimate uncertainty and confidence levels in predictions.

Finally, the assessment should include the limitations of the modeling approach, highlighting any biases or vulnerabilities. Comprehensive quality assurance and control procedures for code verification must also be detailed, ensuring accurate calculations and resolution of anomalies. By addressing these considerations, the credibility and reliability of AI models in regulatory contexts can be thoroughly assessed and documented, facilitating their effective deployment in the drug product life cycle. In the context of AI/ML applications in drug development and clinical trials, advanced QA techniques are transforming testing methodologies to ensure robustness and reliability. However, the FDA draft guidance does not offer actionable steps for incorporating AI-led QA techniques such as autonomous test generation, self-healing mechanisms, acceptance testing, or cognitive test exploration [21, 54].

Autonomous test generation leverages AI to automatically create and adapt test cases, significantly reducing human effort while enhancing test coverage and precision. Self-healing mechanisms streamline QA by enabling systems to identify and resolve test script failures autonomously, ensuring consistent functionality amidst dynamic updates. Cognitive test exploration goes a step further, utilizing AI to perform intelligent exploratory testing by analyzing user behavior, system usage patterns, and emerging risks, allowing the detection of nuanced issues that traditional methods may overlook. Together,

these advanced QA techniques provide a dynamic, adaptive approach to maintaining the integrity of AI/ML models and ensuring patient safety throughout the drug development lifecycle [21, 54].

These methods could streamline clinical data QA and ensure higher reliability standards in regulatory submissions. Moreover, the draft guidance would benefit from examples of successful QA integration into the assessment workflows, particularly in high-risk settings such as manufacturing and clinical trials. These examples can serve as benchmarks for organizations to design and implement robust QA strategies. By incorporating these improvements, the credibility assessment framework will align more closely with modern AI/ML practices, fostering greater trust and reliability in AI-driven systems. Additionally, we recommend that the FDA guidance explicitly incorporate acceptance testing protocols within its risk-based credibility assessment framework, particularly aligning with Step 4. Acceptance testing plays a vital role in establishing the credibility of AI models by validating that they meet predefined requirements and perform reliably within their context of use under regulatory conditions. Including such protocols would strengthen regulatory confidence in AI/ML systems, ensuring that models adhere to safety, efficacy, and quality standards before approval.

6.5. Step 5- Execute the Plan (Improve Phase)

After developing the credibility assessment plan, the next step is to execute it. This involves implementing the planned activities to establish the credibility of the AI model. During this phase, sponsors are encouraged to consult with the FDA to address challenges and refine the credibility assessment activities. The execution phase ensures that the AI model is rigorously tested and evaluated within the context of its intended use, laying the groundwork for ensuring its reliability and effectiveness in regulatory decision-making.

This step involves executing the credibility assessment plan. As discussed in step 4, discussing the plan with the FDA before execution may help (1) set expectations regarding the appropriate credibility assessment activities for the proposed model based on model risk and COU and (2) identify potential challenges and how such challenges can be addressed.

6.6. Step 6- Document the Results (Control Phase)

The results of the credibility assessment plan should be thoroughly documented in a credibility assessment report. This report is intended to provide evidence supporting the credibility of the AI model for its context of use. It should include the findings from steps 1 through 4 and a discussion of any deviations from the original plan. Sponsors should engage with the FDA to determine the appropriate format and timing for submitting the report. Depending on the engagement option, the report may be included as part of a regulatory submission or held for review during inspections. This documentation is critical for demonstrating compliance and establishing the AI model's suitability for its intended purpose.

The results of the credibility assessment plan should be included in a report. For this guidance, this report is referred to as a credibility assessment report. The credibility assessment report is intended to provide information that establishes the credibility of the AI model for the COU and should describe any deviations from the credibility assessment plan as outlined in step 4. During early consultation with the FDA (described in step 4), the sponsor should discuss with the FDA whether, when, and where to submit the credibility assessment report to the Agency. The credibility assessment report may, as applicable, be (1) a self-contained document included as part of a regulatory submission or in a meeting package, depending on the engagement option, or (2) held and made available to FDA on request.

6.7. Step 7- Determine Model Adequacy for the Context of Use (Control and continuous improvement phase)

The final step in the framework is to evaluate the adequacy of the AI model for the defined context of use. Based on the credibility assessment report results, the AI model's

credibility may or may not be considered sufficient for the associated risk level. Several options are available to address this if the credibility is deemed inadequate. Sponsors may incorporate additional evidence, increase the rigor of the assessment, revise the AI model, or adjust the model's context of use. If no adjustments can sufficiently establish credibility, the model may need to be excluded or revised iteratively. This evaluation step ensures that the AI model meets the required standards for its specific regulatory application.

Based on the results documented in the credibility assessment report, a model may or may not be appropriate for the COU. If either the sponsor or the FDA determines that model credibility is not sufficiently established for the model risk, several outcomes are possible:

- The sponsor may downgrade the model influence by incorporating additional types of evidence in conjunction with the evidence from the AI model to answer the question of interest;
- the sponsor may increase the rigor of the credibility assessment activities or augment the model's output by adding additional development data;
- the sponsor may establish appropriate controls to mitigate risk;
- the sponsor may change the modeling approach or
- the sponsor may consider the credibility of the AI model's output inadequate for the COU; therefore, the model's COU would be rejected or revised iteratively.

7. Life Cycle Management

For this guidance, life cycle maintenance refers to managing changes to AI models, incidentally or deliberately, to ensure the model remains fit for use over the drug product life cycle for its COU. Life cycle maintenance of AI models is a set of planned activities to monitor and ensure the model's performance and suitability throughout its life cycle for the COU. Life cycle maintenance of the credibility of AI model outputs is essential because a model's performance can change over time or across deployment environments. While the use of AI to support regulatory decision-making for drugs is typically assessed on locked data and information produced by an AI model at a given point in time, there are instances where the use of AI models extends over the drug product life cycle and life cycle maintenance of the credibility of AI model outputs is critical. For example, life cycle maintenance of the credibility of AI model outputs is essential for the application of AI modeling in the pharmaceutical manufacturing phase of the drug product life cycle (Life cycle maintenance of AI modeling may be necessary during other phases of the drug product life cycle including, but not limited to, the application of AI modeling in the post-marketing phase, as well as AI modeling in the pharmaceutical manufacturing phase, for example.)

AI-based models may be highly sensitive to variations or changes in model inputs, for example, because they are data-driven and can be self-evolving (i.e., capable of autonomously adapting without any human intervention). Model performance metrics should be monitored continuously to ensure the model remains fit and appropriate changes are made to the model, as needed. The level of oversight for a model over its life cycle should be risk-based (i.e., commensurate with the model risk and the COU). Due to the evolving nature of AI models, sponsors should anticipate inherent, model-directed changes and the need to identify and evaluate those changes, as well as any intentional changes to the model over the drug product life cycle.

For life cycle maintenance, a risk-based approach may help sponsors evaluate the impact of a change or changes to the AI model performance [16]. For example, in pharmaceutical manufacturing, changes to the AI model or changes in manufacturing that may impact the performance of the AI model must be evaluated by the manufacturer's change management system within their pharmaceutical quality system (e.g., newly available manufacturing data or information, new signals requiring manual changes in the model, model-directed changes that may impact AI model performance) [55, 56].

The impact of a model change may be determined based on factors such as model risk and change in model performance. Depending on the extent of the change and its impact on model performance, some steps in the credibility assessment plan may need to be re-executed, including retraining and retesting the model for the COU. Additionally, depending on the impact of model change (i.e., if the model change impacts model performance), the change should be reported to the Agency per regulatory requirements (For example, as appropriate for the application type, such update would generally be made as a post-approval change by section 506A of the FD&C Act and 21 CFR 314.70 (for human drugs) [57], 21 CFR 601.12 (for human biological products) [58], or 21 CFR 514.8 (for animal drugs) [59]. The mechanism for post-approval notification of changes to models can be determined based on the impact of the change on the model's performance and product quality.

In general, detailed plans about life cycle maintenance (e.g., model performance metrics, the risk-based frequency for monitoring model performance, and triggers for model retesting) should be made available for review as a component of the manufacturing site's pharmaceutical quality system, with a summary included in the marketing application for any product or process-specific models, under regulatory requirements (See 21 CFR 314.50 and 601.2. [60, 61]).

FDA recommends that the level of detail regarding the life cycle maintenance of the AI model be commensurate with model risk. Sponsors may also choose to use tools outlined in the ICH guidance for industry Q12 Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management (May 2021) [62], such as established conditions and comparability protocols (referred to as post-approval change management plans), which leverages increased product and process knowledge. Sponsors may propose model-related elements to be considered established conditions, along with a plan to manage changes to these established conditions over the drug product life cycle. By including such plans in the marketing application, sponsors may prospectively obtain input from the Agency regarding managing such changes, including which changes would not require submission to the Agency before making modifications.

The iterative and autonomous nature of AI/ML-enabled tools demands a shift from traditional static regulatory models to a Total Product Lifecycle Assurance (TPLA) framework. This approach emphasizes continuous monitoring and frequent reviews to ensure these technologies remain safe, effective, and aligned with clinical needs. A robust TPLA would include continuous assurance protocols, allowing for regular evaluation and updates of adaptive AI systems. Despite these advancements, regulatory authorities face fundamental questions about when modifications to self-learning systems should trigger new premarket reviews or conformity assessments. For example, the threshold for defining a "significant change" to the system remains ambiguous across regulatory frameworks, including FDA and European Union standards.

The FDA's 2019 [63] for AI/ML-based SaMD (Software as a Medical Device) highlights the need for a Predetermined Change Control Plan (PCCP) during the initial regulatory review. This plan anticipates potential software modifications and provides a structured methodology for ensuring post-modification safety and effectiveness. However, the FDA also recognizes that additional regulatory reviews may be necessary depending on the scope of these modifications. In contrast, the EU allows initial conformity assessments to approve self-learning mechanisms, limiting further reviews unless the system undergoes unsupervised modifications. The EU White Paper underscores the importance of operational constraints to ensure AI systems cannot autonomously override safety mechanisms without human intervention or additional certification [64] (Table 2).

Table 2. FDA and EU perspectives on AI in drug development regulatory decision-making.

Regulatory Framework	FDA Draft Guidance (January 2025), [16, 19]	EMA Reflection Paper (September 2024), [64, 19]
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Focus	Risk-based credibility assessment for AI/ML models in drug development	Standardization and harmonization of AI/ML use in pharmaceuticals
Patient Safety	Ensuring safety via continuous AI oversight in drug development (i.e., high risk situation)	Ensuring patient safety and efficacy with uniform regulations (i.e., high risk situation)
Data Quality	Emphasizes general aspects of reliable data for safe and effective AI models	Focuses on data quality in the context of cross-border harmonization and consistent standards (SOP/GxP) across the EU, transparency, data integrity and ethical aspects of trustworthy data as “Raw Material” qualification for AI/ML models, and data protection compliance
Model Development	Emphasizes on general aspects of model development and 7-steps risk assessment	Encourages usage of mathematical algorithms with parameters arranged in an architecture that allows learning of patterns from training data, and overfitting assessment, especially in clinical settings, performance assessment including confusion matrix, interpretability, explainability
Model Validation	Clear step by step guidance on general aspects of validating AI models	Stresses the importance of model validation and rigorous testing to ensure AI reliability
Life Cycle Management	Acknowledged life-cycle management for AI driven drug development via early regulatory engagement	Emphasis on risk assessment of AI- models for full life cycle management of medicinal product

As noted previously, the FDA strongly encourages sponsors and other interested parties to engage early with the FDA to set expectations regarding the appropriate credibility assessment activities for the proposed model based on model risk and COU and help identify potential challenges and how such challenges may be addressed.

Depending on how the sponsor or other interested parties intend to use the AI model, various options can be used to engage with the Agency. To discuss the use of AI in connection with a specific development program, sponsors may request an appropriate formal meeting (e.g., Initial Regulations at 21 CFR_312.82 [47, 65, 66]).

The guideline lists various other engagement options depending on the intended use of the AI model. Where the meeting request covers a specific development program under an investigational new drug application (IND) or a pre-IND, sponsors should include the IND or pre-IND number and notify the relevant review team of the meeting request.

8. Conclusion

The FDA’s draft guidance represents a significant milestone in regulating AI in drug and biological product development. Its emphasis on risk-based assessments, early stakeholder engagement, and model transparency underscores the agency’s commitment to fostering innovation while maintaining rigorous regulatory oversight. However, addressing several key gaps and challenges will enhance the guidance’s utility and ability to

support AI's safe, ethical, and equitable integration into regulatory decision-making. In essence, the technopolitical dynamics of AI are evident in narratives emphasizing its transformative power and disruptive potential. Regulators and policymakers can amplify the ethical and trust-related pressures on AI-driven systems by navigating the complexity of safety, efficacy, and ethical considerations while fostering confidence in technology. Transparent risk communication, robust oversight mechanisms, and inclusive stakeholder engagement are critical to addressing concerns and legitimizing the deployment of AI-driven solutions in sensitive high-risk domains like healthcare. Building trust involves mitigating risks, demonstrating accountability, and aligning system outcomes with societal values.

To strengthen its impact, the guidance expands its scope to provide actionable strategies for mitigating bias in AI models and addressing resource disparities that may hinder more minor sponsors. Incorporating advanced methodologies, such as pragmatic trials, Bayesian modeling, and EHR interoperability, will improve AI models' robustness and applicability in real-world settings. Additionally, the guidance should include specific recommendations for continuous assurance protocols, ensuring real-time oversight and performance monitoring of adaptive AI systems, especially in high-stakes applications like patient care. Given the evolving nature of AI and as shown by the 7-step model consideration, lifecycle maintenance is crucial for ensuring sustained model performance. Sponsors must plan for ongoing monitoring of AI outputs, retraining and retesting in response to data drift or significant model changes, and compliance with regulatory requirements for post-approval changes.

The integration of RWD and RWE across the entire drug lifecycle, from early development to post-market surveillance, is crucial. Expanding and aligning AI and RWE frameworks would bridge existing gaps, enabling a unified approach to regulatory decision-making. Furthermore, ethical oversight must be strengthened by requiring explicit bias assessments, human-in-the-loop safeguards, and alignment with international standards, such as the G7's principles, to ensure consistency and equity on a global scale.

The draft guidance must also define clear metrics and methodologies for evaluating data quality and generalizability during initial product development. Addressing these aspects will prevent unintended biases and ensure AI models are reliable across diverse populations and geographies. Harmonizing the FDA's framework with cross-border regulatory efforts will help establish globally aligned standards for adaptive and iterative AI technologies, facilitating smoother adoption and implementation for multinational sponsors.

Finally, incorporating patient perspectives into the framework ensures patient-centered AI applications. Addressing patient preferences, accessibility challenges, and bias mitigation will enhance trust and engagement while improving equity and innovation in drug development. By fostering collaboration with stakeholders, including HTA bodies, payers, and international regulators, the FDA can refine its guidance to create a truly comprehensive and forward-looking regulatory tool. Moreover, by leveraging adaptive AI-driven QA strategies, the challenges in traditional QA and acceptance testing can be overcome, resulting in more robust, scalable, and reliable model risk assessment processes that align with the increasing complexity of modern software and AI/ML systems. In AI-driven QA, acceptance testing could involve validating model predictions, assessing algorithm robustness against diverse datasets, and confirming explainability and transparency of results. This phase also helps mitigate risks by identifying potential safety, efficacy, or quality issues. As in the FDA guidance, regulatory decision-making is closely tied to steps like developing a plan to establish credibility (Step 4) and ensuring the AI model meets credibility and performance benchmarks before approval. We recommend that the FDA consider further incorporating statistical predictive methods such as Bayesian along with RWE in future guidance to adapt to the evolving landscape of AI in drug development. This will ensure that regulatory frameworks remain flexible, data-driven, and reflective of real-world patient outcomes.

Stakeholder feedback during the comment period will be vital in refining these areas. By incorporating the recommendations discussed, the FDA will advance its regulatory science and establish a robust framework that supports innovative, ethical, and patient-centric AI applications in drug development, ensuring safety, quality, and equity remain at the forefront of its mission.

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